EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 1, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-03 **Date:** July 23, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. How many rows appear in a truth table for each of these compound propositions? You do not need to construct the truth tables, only compute the row counts. **[CO1, C2] [Mark: 5]**

(a) $p \wedge q$

(b) $(p \leftrightarrow q) \wedge (q \rightarrow r)$

(c) $(p \leftrightarrow q) \vee (r \leftrightarrow s)$

(d) $(p \vee \neg q) \wedge (\neg r \vee s)$

(e) $(p \wedge q \wedge r) \vee (s \wedge t)$

2. Use a truth table to verify the **absorption law** in propositional logic:

$$p \vee (p \wedge q) \equiv p$$

Construct a truth table for both sides of the equivalence and show that they are logically equivalent.

[CO1, C2, Mark: 5]

EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 1, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-09 **Date:** July 24, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted toward the final grade)
Time: 20 Minutes

Note: There are **TWO** questions. Answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Let p and q be the propositions:
p: It is below freezing.
q: It is snowing.

Write the following statements using p and q, and logical connectives (including negations):

a) It is below freezing and snowing.

b) It is below freezing but not snowing.

c) It is not below freezing and it is not snowing.

d) It is either snowing or below freezing (or both).

e) If it is below freezing, it is also snowing.

[CO1, C2] [Mark: 5]

2. Use a truth table to verify the **Distributive Law** in propositional logic. Construct a truth table for both sides of the equivalence and show that they are logically equivalent.

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

[CO1, C2] [Mark: 5]

EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 1, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-10 **Date:** July 24, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted toward the final grade)
Time: 20 Minutes

Note: There are **THREE** questions. Answer **ALL** of them. Course Outcome (CO), Cognitive Level, and marks for each question are mentioned in the right margin.

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1. Let p, q and r be the propositions:
p: You have the flu.
q: You miss the final examination.
r: You pass the course.
Write the following statements using p, q, and r and logical connectives (including negations):

a) If you have the flu, you miss the final examination.

b) You do not miss the final examination if and only if you pass the course.

c) If you miss the final examination, you do not pass the course.

[CO1, C2] [Mark: 3]

2. How many rows appear in a truth table for each of these compound propositions? You do not need to construct the truth tables, only compute the row counts.

(a) $(p \wedge q) \rightarrow (r \vee s)$

(b) $(p \leftrightarrow (q \vee r)) \wedge (s \rightarrow (t \vee u))$

[CO1, C2] [Mark: 2]

3. Use a truth table to verify the following **De Morgan's Law** in propositional logic. Construct a truth table for both sides of the equivalence and show that they are logically equivalent.

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

[CO1, C2] [Mark: 5]